

Serum Neurofilament Light Chain and Cardiovascular Outcomes in Patients With Atrial Fibrillation

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 Supplemental content

IMPORTANCE Serum neurofilament light chain (sNfL) is a blood-based biomarker initially investigated for neuronal injury and is elevated in patients with atrial fibrillation (AF). Whether sNfL is associated with other adverse outcomes in this population is less well studied.

OBJECTIVE To determine the association between sNfL and cardiovascular outcomes in patients with AF.

DESIGN, SETTING, AND PARTICIPANTS The Swiss Atrial Fibrillation Cohort (SWISS-AF) is a prospective, multicenter, observational cohort study. The present analysis included patients enrolled from April 2014 through August 2017, with follow-up until April 23, 2025. Data analysis was conducted in June 2025. SWISS-AF enrolled inpatients and outpatients across 14 secondary or tertiary care centers in Switzerland. This analysis included 2311 of 2415 (95.7%) patients with documented AF in SWISS-AF after the exclusion of 76 patients without sNfL measurements and 28 patients with missing follow-up data.

EXPOSURES Concentrations of sNfL were measured from baseline serum samples using an ultrasensitive single-molecule array assay.

MAIN OUTCOMES AND MEASURES The primary outcome was major vascular events (MVEs), defined as a composite of cardiovascular death, nonfatal stroke, and nonfatal myocardial infarction. Other outcomes included the individual components of MVEs, heart failure-related hospitalization, and all-cause mortality.

RESULTS Of 2311 included patients, mean (SD) age was 73.2 (8.5) years, and 1683 (73%) were male. Over a median (IQR) follow-up of 8.0 (5.2-9.0) years, there were 665 first MVE events, corresponding to an incidence per 100 patient-years of 4.4 (95% CI, 4.1-4.8). Every doubling of sNfL concentration was associated with an adjusted hazard ratio (aHR) for MVEs of 1.35 (95% CI, 1.22-1.50; $P < .001$). Increasing sNfL levels were associated with nonfatal stroke (aHR, 1.31; 95% CI, 1.09-1.57; $P = .004$), cardiovascular death (aHR, 1.36; 95% CI, 1.20-1.54; $P < .001$), heart failure-related hospitalization (aHR, 1.25; 95% CI, 1.11-1.41; $P < .001$), and all-cause mortality (aHR, 1.41; 95% CI, 1.27-1.56; $P < .001$) but not myocardial infarction (aHR, 1.04; 95% CI, 0.81 to 1.34; $P = .76$).

CONCLUSIONS AND RELEVANCE After multivariable adjustment, sNfL levels were associated with a broad range of adverse cardiovascular events and mortality in patients with AF. Serum NfL may serve as a biomarker of cardiovascular risk in patients with AF.

JAMA Cardiol. doi:10.1001/jamacardio.2026.0922
Published online March 29, 2026.

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Despite major advances in how we treat patients with atrial fibrillation (AF), they remain at high risk of adverse cardiovascular outcomes, including heart failure (HF), myocardial infarction, and cardiovascular death.¹⁻⁵ These persistent risks highlight the importance of identifying mechanisms associated with these complications.

Serum neurofilament light chain (sNfL) is a biomarker released into cerebrospinal fluid during neuronal injury and subsequently released into the peripheral circulation.^{6,7} sNfL levels are elevated in patients with AF compared to healthy control individuals, even among those without a history of major cerebrovascular events such as stroke or transient ischemic attack (TIA).⁸ A previous analysis also found that higher sNfL levels were associated not only with magnetic resonance imaging-based markers of subclinical cerebral injury (eg, white matter lesion volume, increased cerebral infarct volume), but also with diabetes, peripheral artery disease, HF, and blood pressure.⁹ These findings suggest that elevated sNfL levels may reflect a broader cardiovascular risk profile beyond the brain.

Small observational studies have found associations between sNfL and adverse cardiovascular outcomes in non-AF populations. They found that increasing sNfL levels were associated with cardiovascular death in patients with suspected coronary artery disease, risk of HF hospitalization in patients with reduced ejection fractions, and all-cause mortality in several populations.¹⁰⁻¹⁴ However, the limited sample sizes and number of outcome events in these studies restrict the robustness and generalizability of these associations.

Patients with AF may be of particular importance in this regard, given several unique pathways of neurovascular and systemic vascular injury that may link these entities, such as cerebral hypoperfusion, microembolism, and atrial myopathy.^{15,16} Two previous publications of overlapping populations have investigated the association of sNfL with cerebrovascular and cardiovascular outcomes in patients with AF, but were limited by short follow-up duration, low event numbers, and most importantly were conducted many years ago in patients not receiving oral anticoagulation medication.^{17,18} It is therefore uncertain whether the results of these studies are applicable to contemporary patients who often receive oral anticoagulation and other state-of-the-art treatments. Therefore, the aim of this study was to evaluate the associations between circulating sNfL concentrations and cardiovascular outcomes in a large, contemporary cohort of patients with AF followed up for an average of 8 years and most of whom were receiving oral anticoagulation medication.

Methods

Study Design and Population

We conducted an analysis of the Swiss Atrial Fibrillation Cohort (SWISS-AF, [NCT02105844](#)), a prospective, multicenter, observational cohort study at 14 centers in Switzerland. Consecutive patients were included if aged 65 years or older and with a previous diagnosis of AF, either as paroxysmal (AF lasting <7 days not requiring cardioversion), persistent (sustained AF ≥7 days and/or requiring cardioversion), or perma-

Key Points

Question Are serum levels of neurofilament light chain (sNfL) associated with adverse cardiovascular outcomes in patients with atrial fibrillation (AF)?

Finding In this cohort study of 2311 patients with AF, increasing sNfL levels were significantly associated with an increased risk of major vascular events, nonfatal stroke, cardiovascular mortality, heart failure hospitalization, and all-cause mortality.

Message The findings in this study suggest that sNfL, a specific marker of neuronal injury, may be useful in identifying patients with AF who are at heightened cardiovascular risk, providing novel insights beyond traditional cardiovascular risk factors.

nent (failed or unattempted cardioversion). Exclusion criteria for the main study were inability to provide informed consent, secondary forms of AF, and acute illness within 4 weeks of enrollment. A small group of patients younger than 65 years (around 10%) was included to investigate AF in the working population. A total of 2415 patients were enrolled from 2014 to 2017. For this analysis, we excluded 76 (3.1%) patients without sNfL measurement (no or insufficient blood sample) and 28 (1.2%) patients without follow-up data.

Patients were followed up from cohort inclusion until death, loss to follow-up, or date of latest data extraction (April 23, 2025), whichever came first. Standardized data collection forms were used at baseline and yearly visits to collect detailed clinical and sociodemographic data, treatments, interventions, and intercurrent adverse events. The study protocol was approved by local ethics committees, and informed written consent was obtained from each participant. Further information about the methodology for this cohort has been published previously.^{19,20} This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline.

sNfL Measurement

Serum samples were collected at baseline from included patients.¹⁹ Samples were centrifuged and stored at -80 °C in a centralized biobank. Concentrations of sNfL were measured in duplicate using a previously described ultrasensitive single-molecule array assay developed in-house.^{9,21} The interassay coefficients of variation were 10%, 12%, and 5% for low, medium, and high concentration quality control serum samples, respectively. The mean intraassay coefficient of variation was 5%. Individuals measuring sNfL were blinded to clinical and outcome data. The dynamic range of the assay was 1 pg/mL to 2000 pg/mL. There were no samples with values above the upper limit of sensitivity. Sixteen patients had sNfL concentrations below the lower limit of detection and were imputed to have a value of 0.5 pg/mL (lower limit of detection/2).

Outcomes

All major cardiovascular outcomes were adjudicated. The primary outcome for this analysis was major vascular events (MVEs), defined as a composite of cardiovascular death, nonfatal stroke (ischemic or hemorrhagic), and nonfatal myocardial infarction. Secondary outcomes included the individual

components of the primary composite outcome, HF-related hospitalizations, and all-cause mortality. The definitions of all end points are shown in eTable 1 in [Supplement 1](#).

Statistical Analysis

Baseline characteristics were stratified by sNfL quartiles. Normally distributed continuous data are presented as means with SDs, nonnormally distributed continuous data as medians with IQRs, and categorical data as numbers with percentages. Continuous variables were compared using analysis of variance if normally distributed and the Kruskal-Wallis test if not normally distributed. Categorical variables were compared using χ^2 tests.

Event rates for each outcome were estimated using the Kaplan-Meier method, stratified by sNfL quartiles, with patients censored at their last available follow-up. To assess the associations between sNfL and individual outcomes, we performed multivariable Cox regression analyses. Since sNfL levels were right skewed, we log-transformed the values. Models were first adjusted for age and sex, and then for a predefined set of covariates (based on literature review and expert advice), including age, sex, body mass index, history of HF, hypertension, coronary artery disease, diabetes, prior stroke or TIA, peripheral arterial disease, glomerular filtration rate, type of AF, baseline electrocardiogram rhythm, and smoking status. To allow for nonlinear associations, log-transformed sNfL was first entered into the model as a restricted cubic spline with 3 knots placed at the 10th, 50th, and 90th percentiles. Since the associations between sNfL and the log hazard appeared to be close to linear, we proceeded assuming linear relationships (eFigure 1 in [Supplement 1](#)). We report hazard ratios (HRs) and 95% CIs per doubling of nontransformed sNfL levels.

For the primary outcome of MVE, we evaluated several subgroup analyses: (1) type of AF (paroxysmal vs persistent vs permanent), (2) anticoagulation medication status at baseline, and (3) prior stroke or TIA. We hypothesized that the association between sNfL and MVE would be stronger in patients not receiving anticoagulation medication at baseline, patients without prior stroke or TIA, and patients with permanent AF at baseline. For each subgroup analysis, an interaction term between sNfL and the subgroup variables was entered into the models and fit on the whole population. For the outcome of HF-related hospitalization, we completed a sensitivity analysis excluding patients with prior history of HF. Additionally, for all prespecified outcomes, we completed a sensitivity analysis of the most adjusted multivariable model, additionally adjusting for biomarkers known to be associated with cardiovascular outcomes in AF, including N-terminal pro-B-type natriuretic peptide (NT-proBNP), C-reactive protein, and troponin T. In this sensitivity analysis, an additional 133 patients were excluded due to missing biomarker data. To assess the incremental prognostic value of sNfL, we evaluated changes in model performance after adding sNfL to the ABC-AF risk score.²² The ABC-AF risk score incorporates age, prior stroke, NT-proBNP, and high-sensitivity troponin T and was derived per patient using the published risk equation.^{22,23} The linear predictor was recalibrated to estimate 8-year risk (based on median follow-up duration in our cohort) of MVEs and

stroke, and the incremental prognostic value of sNfL was quantified using changes in the C statistic and continuous integrated discrimination improvement and net reclassification improvement. All analyses and visualizations were done with Stata version 18 (StataCorp). All tests were 2-sided, with $P < .05$ considered statistically significant.

Results

Among the 2311 patients, the mean (SD) age was 73.2 (8.5) years; 1683 (73%) were male and 628 (27%) were female. The median (IQR) sNfL was 41.8 (28.6-64.9) pg/mL, with a range from 0.5 pg/mL to 910.8 pg/mL.

Patient characteristics across quartiles of sNfL levels are shown in [Table 1](#). Compared to patients in the lower sNfL quartiles, those in the highest quartile were older, had lower body mass index, and were less frequently active smokers. Patients in the highest quartile also had higher CHA2DS2-VASc scores with a higher prevalence of hypertension, coronary artery disease, diabetes, prior stroke or TIA, HF, and peripheral arterial disease.

MVEs

The median (IQR) follow-up was 7.97 (5.19-9.00) years, during which the composite outcome of MVE occurred in 665 patients, with an incidence rate of 4.42 (95% CI, 4.10-4.77) per 100 patient-years, ranging from 1.89 in the lowest quartile up to 9.47 in the higher quartile ([Table 2](#)). Event-free survival curves for MVE stratified by sNfL quartiles are shown in [Figure 1A](#). At 8 years, the incidence of the primary outcome across increasing sNfL quartiles was 14.3%, 22.3%, 31.5%, and 55.1%, respectively. In the multivariable adjusted model, every doubling of sNfL concentration was associated with an aHR of 1.35 (95% CI, 1.22-1.50; $P < .001$) for the primary outcome.

Secondary Outcomes

Event-free survival curves for secondary outcomes are shown in [Figure 1B-D](#) and [eFigure 2](#) in [Supplement 1](#). Incidence rates stratified by sNfL quartiles and associations between sNfL concentration and outcomes are shown in [Tables 2 and 3](#). Every doubling of sNfL concentration was associated with aHR for nonfatal stroke of 1.31 (95% CI, 1.09-1.57; $P = .004$), an aHR for nonfatal myocardial infarction of 1.04 (95% CI, 0.81-1.34; $P = .76$), an aHR for cardiovascular death of 1.36 (95% CI, 1.20-1.54; $P < .001$), an aHR for HF-related hospitalization of 1.25 (95% CI, 1.11-1.41; $P < .001$), and an aHR for all-cause mortality of 1.41 (95% CI, 1.27-1.56; $P < .001$). In a sensitivity analysis only including patients without a history of HF, increased sNfL remained significantly associated with HF-related hospitalization (aHR, 1.38; 95% CI, 1.18-1.61; $P < .001$).

Subgroup Analyses

Prespecified subgroup analyses for the outcome of MVE are shown in [Figure 2](#). As hypothesized, the association between increasing sNfL and MVE was stronger in patients not taking anticoagulation medication at baseline (HR, 1.82; 95% CI, 1.41-2.35) than those taking anticoagulation medication at base-

Table 1. Baseline Characteristics of the Population, Stratified by Serum Neurofilament Light Chain (sNfL) Quartile (Q)

Characteristic	sNfL quartile				P value ^a
	Q1 (<1.0-28.6 pg/mL)	Q2 (28.6-41.8 pg/mL)	Q3 (41.8-64.9 pg/mL)	Q4 (64.9-910.8 pg/mL)	
No. of patients	578	579	577	577	
Age, median (IQR), y	67.2 (60.4-71.7)	72.5 (68.2-76.7)	76.0 (71.4-80.4)	78.6 (73.8-83.0)	<.001
Sex, No. (%)					
Male	452 (78.2)	424 (73.2)	391 (67.8)	416 (72.1)	.001
Female	126 (21.8)	155 (26.8)	186 (32.2)	161 (27.9)	
BMI, median (IQR)	28.0 (25.3-31.7)	27.4 (24.6-30.7)	26.3 (24.3-29.7)	26.3 (23.6-29.4)	<.001
Type of AF, No. (%)					
Paroxysmal	284 (49.1)	254 (43.9)	257 (44.5)	231 (40.0)	<.001
Permanent	104 (18.0)	146 (25.2)	141 (24.4)	185 (32.1)	
Persistent	190 (32.9)	179 (30.9)	179 (31.0)	161 (27.9)	
Rhythm at blood collection, No. (%)					
Sinus rhythm	397 (64.9)	332 (53.0)	339 (51.3)	294 (40.8)	<.001
Atrial fibrillation	181 (35.1)	247 (47.0)	238 (48.7)	283 (59.2)	
Smoking, No. (%)					
Never	228 (39.5)	253 (43.8)	280 (48.5)	251 (43.5)	.005
Former	291 (50.4)	278 (48.1)	263 (45.6)	295 (51.1)	
Active	58 (10.1)	47 (8.1)	34 (5.9)	31 (5.4)	
Alcohol, median (IQR), g/d	0.6 (0.1-1.4)	0.6 (0.1-1.3)	0.6 (0.1-1.3)	0.3 (0.1-1.1)	<.001
CHA2DS2-VASc score, mean (SD)	2.3 (1.5)	3.2 (1.5)	3.8 (1.5)	4.4 (1.5)	<.001
eGFR, median (IQR), mL/min	72.8 (62.6-83.3)	61.7 (54.2-73.1)	55.8 (45.1-65.0)	44.5 (31.8-56.2)	<.001
NT-proBNP, median (IQR), pg/mL	277 (102-705)	539 (197-1237)	730 (303-1525)	1363 (619-2584)	<.001
C-reactive protein, median (IQR), mg/L	1.3 (0.7-2.7)	1.6 (0.8-3.6)	1.7 (0.8-4.2)	2.5 (1.1-5.7)	<.001
hs-Troponin T, median (IQR), pg/mL	9.6 (7.1-13.7)	12.8 (9.2-18.2)	15.7 (11.2-23.5)	26.7 (17.0-42.0)	<.001
Comorbidities					
Hypertension, No. (%)	359 (62.1)	385 (66.5)	417 (72.3)	459 (79.5)	<.001
Coronary artery disease, No. (%)	118 (20.4)	170 (29.4)	188 (32.6)	231 (40.0)	<.001
Diabetes, No. (%)	52 (9.0)	89 (15.4)	96 (16.6)	166 (28.8)	<.001
Prior stroke/TIA, No. (%)	71 (12.3)	107 (18.5)	133 (23.1)	152 (26.4)	<.001
Heart failure, No. (%)	80 (13.8)	131 (22.7)	162 (28.1)	231 (40.0)	<.001
Myocardial infarction, No. (%)	59 (10.2)	85 (14.7)	103 (17.9)	131 (22.7)	<.001
Peripheral arterial disease, No. (%)	19 (3.3)	41 (7.1)	38 (6.6)	87 (15.1)	<.001
Systemic embolism, No. (%)	13 (2.2)	30 (5.2)	38 (6.6)	41 (7.1)	<.001
Kidney failure, No. (%)	26 (4.5)	70 (12.1)	119 (20.7)	274 (47.5)	<.001
Medications					
Anticoagulant, No. (%)	506 (87.5)	529 (91.4)	525 (91.0)	528 (91.5)	.07
Antiplatelet, No. (%)	73 (12.6)	118 (20.4)	115 (20.0)	149 (25.9)	<.001
β-Blocker, No. (%)	395 (68.3)	412 (71.2)	397 (68.8)	425 (73.7)	.17
Statin, No. (%)	247 (42.7)	283 (48.9)	296 (51.3)	326 (56.5)	<.001
RAAS inhibitor, No. (%)	322 (55.7)	357 (61.7)	368 (63.8)	405 (70.2)	<.001
Mineralocorticoid receptor antagonist, No. (%)	51 (8.8)	60 (10.4)	77 (13.3)	116 (20.1)	<.001

Abbreviations: AF, atrial fibrillation; BMI, body mass index (calculated as weight in kilograms divided by height in meters squared); eGFR, estimated glomerular filtration rate; hs-troponin T, high-sensitivity troponin T; NT-proBNP, N-terminal pro-B-type natriuretic peptide; RAAS, renin-angiotensin-aldosterone system; TIA, transient ischemic attack.

^a P values reported for Pearson χ^2 test for categorical variable, ANOVA test for normally distributed continuous variables, and Kruskal-Wallis test for non-normally distributed continuous variables.

line (HR, 1.31; 95% CI, 1.18-1.47) (interaction $P = .02$). As hypothesized, the association between sNfL and MVE was stronger in patients with no prior stroke or TIA (HR, 1.48; 95% CI, 1.31-1.69) than in patients with prior stroke or TIA (HR, 1.17; 95% CI, 0.99-1.38) (interaction $P = .02$). The association between sNfL and MVE did not differ across type of AF at baseline. An exploratory post hoc subgroup analysis found no significant difference in effect size between patients in sinus

rhythm (aHR, 1.35; 95% CI, 1.15-1.59; $P < .001$) vs AF (aHR, 1.35; 95% CI, 1.20-1.53; $P < .001$) (interaction $P = .97$) at baseline.

Sensitivity Analysis Adjusting for Other Biomarkers

We conducted a sensitivity analysis to assess the association between sNfL and all outcomes, additionally adjusting for other biomarkers including NT-proBNP, C-reactive protein, and troponin T. eTable 2 in [Supplement 1](#) summarizes the results of

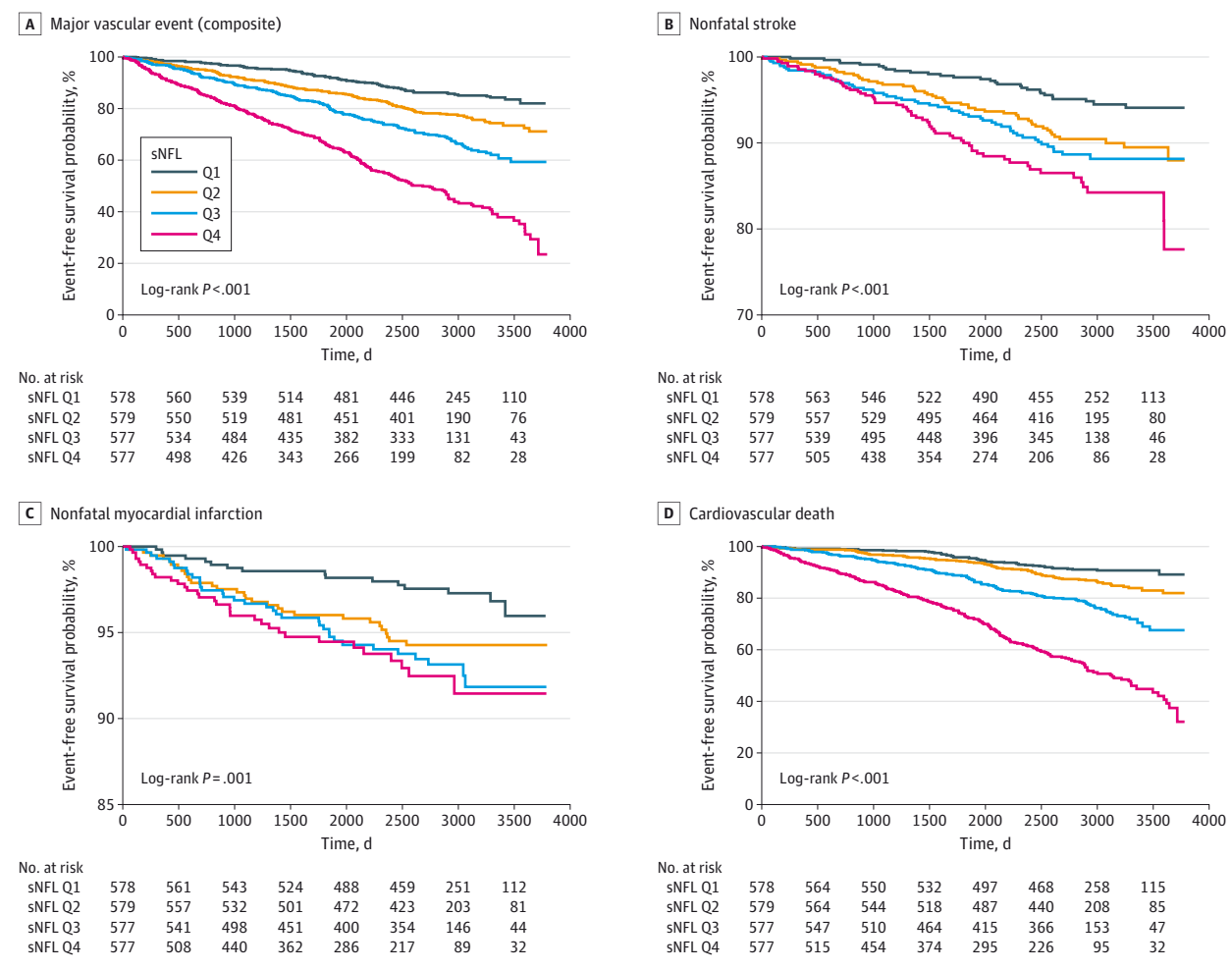
Table 2. Incidence Rates for Outcomes in Total Population and by Serum Neurofilament Light Chain (sNfL) Quartile (Q)

Outcome	sNfL quartile								Total (N = 2311)	
	Q1 (n = 578)		Q2 (n = 579)		Q3 (n = 577)		Q4 (n = 577)			
	No. of events	Incidence rate (95% CI) ^a	No. of events	Incidence rate (95% CI) ^a	No. of events	Incidence rate (95% CI) ^a	No. of events	Incidence rate (95% CI) ^a	No. of events	Incidence rate (95% CI) ^a
Major vascular event	83	1.89 (1.52-2.34)	130	3.19 (2.69-3.79)	175	4.80 (4.14-5.57)	277	9.47 (8.42-10.66)	665	4.42 (4.10-4.77)
Stroke	28	0.62 (0.43-0.91)	51	1.22 (0.93-1.61)	54	1.45 (1.11-1.89)	61	2.04 (1.59-2.62)	194	1.26 (1.10-1.46)
Myocardial infarction	16	0.36 (0.22-0.59)	30	0.71 (0.50-1.02)	35	0.93 (0.67-1.29)	33	1.08 (0.77-1.52)	114	0.74 (0.61-0.88)
Cardiovascular death	50	1.11 (0.84-1.46)	77	1.78 (1.43-2.23)	125	3.23 (2.71-3.85)	238	7.60 (6.70-8.63)	490	3.09 (2.83-3.38)
HF-related hospitalization	63	1.46 (1.14-1.86)	110	2.76 (2.29-3.33)	148	4.30 (3.66-5.05)	231	8.98 (7.90-10.22)	552	3.85 (3.55-4.19)
All-cause mortality	75	1.66 (1.33-2.08)	130	3.01 (2.53-3.57)	197	5.09 (4.43-5.85)	354	11.31 (10.19-12.55)	756	4.77 (4.45-5.13)

Abbreviation: HF, heart failure.

^a Incidence reported as rate per 100 patient-years of follow-up.

Figure 1. Cumulative Event-Free Survival Curves for Outcomes Stratified by Serum Neurofilament Light Chain (sNfL) Quartiles (Qs)



the sensitivity analyses. The associations remained significant between increasing sNfL and MVE, nonfatal stroke, cardiovascular death, HF-related hospitalization, and all-cause mortality.

Incremental Prognostic Value of sNfL

For MVE, addition of sNfL to the ABC-AF score increased the C statistic from 0.696 to 0.713, with a Δ C statistic of 0.017 (95% CI, 0.008 to 0.029; $P < .001$), an integrated discrimination im-

Table 3. Association Between Serum Neurofilament Light Chain Levels and Outcomes, by Model

Outcome	Model		Age and sex adjusted		Multivariable adjusted ^a	
	Unadjusted					
	HR (95% CI)	P value	HR (95% CI)	P value	HR (95% CI)	P value
Major vascular event	1.74 (1.63-1.87)	<.001	1.48 (1.36-1.60)	<.001	1.35 (1.22-1.50)	<.001
Nonfatal stroke	1.53 (1.34-1.76)	<.001	1.33 (1.13-1.56)	.001	1.31 (1.09-1.57)	.004
Nonfatal myocardial infarction ^b	1.36 (1.13-1.63)	.001	1.17 (0.95-1.46)	.142	1.04 (0.81-1.34)	.76
Cardiovascular death	1.88 (1.75-2.03)	<.001	1.54 (1.41-1.69)	<.001	1.36 (1.20-1.54)	<.001
HF-related hospitalization	1.75 (1.62-1.88)	<.001	1.51 (1.38-1.65)	<.001	1.25 (1.11-1.41)	<.001
All-cause mortality	1.88 (1.76-1.99)	<.001	1.57 (1.46-1.69)	<.001	1.41 (1.27-1.56)	<.001

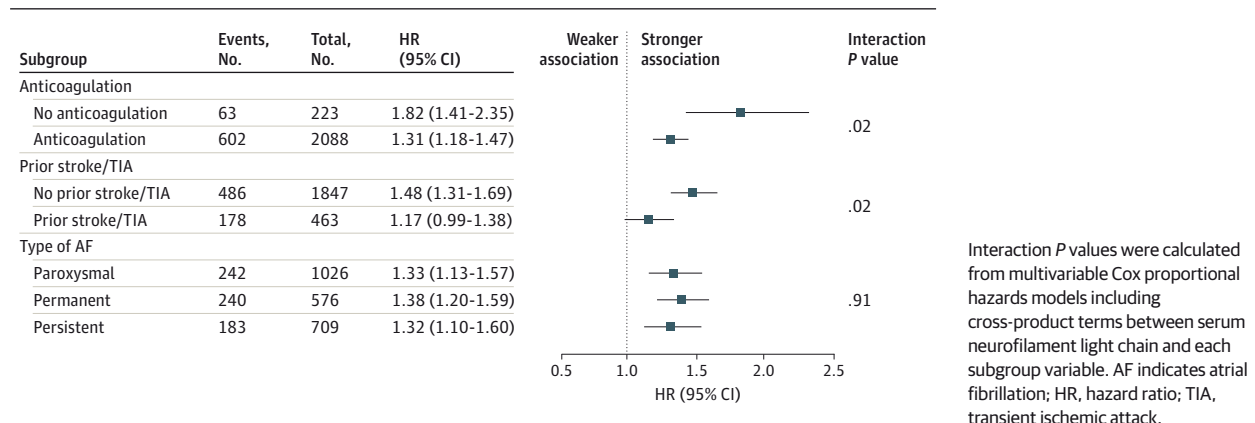
Abbreviations: HF, heart failure; HR, hazard ratio.

^a Adjusted for age, sex, body mass index, heart failure, hypertension, coronary artery disease, diabetes, prior stroke or transient ischemic attack, peripheral arterial disease, glomerular filtration rate, type of atrial fibrillation, baseline electrocardiogram rhythm, and smoking status. The multivariable-adjusted

model was a complete-case analysis based on 2271 patients, after exclusion of 40 (1.73%) with missing covariate data.

^b For the outcome of myocardial infarction, due to the number of events, the following covariates were removed in the multivariable adjusted model: type of atrial fibrillation, baseline electrocardiogram rhythm, and smoking status.

Figure 2. Forest Plot Showing Prespecified Subgroup Analyses for Major Vascular Event Outcome



provement of 0.023 ($P = .002$), and a net reclassification improvement of 0.182 ($P < .001$) (eTable 3 in Supplement 1). For stroke, the addition of sNfL increased the C statistic from 0.675 to 0.686, with a ΔC statistic of 0.013 (95% CI, -0.000 to 0.032 ; $P = .07$), an integrated discrimination improvement of 0.003 ($P = .13$), and a net reclassification improvement of 0.190 ($P = .01$) (eTable 3 in Supplement 1).

Discussion

In this analysis of the SWISS-AF cohort, increasing sNfL levels were significantly associated with a broad range of adverse outcomes, including MVEs, nonfatal stroke, cardiovascular death, HF-related hospitalization, and all-cause mortality. The associations were attenuated after comprehensive multivariable adjustment but remained significant for all outcomes except myocardial infarction. The associations and effect sizes remained consistent after additional adjustment for other biomarkers including NT-proBNP, troponin T, and C-reactive protein.

This is one of the first large long-term prospective studies to show associations between sNfL levels and adverse cardiovascular outcomes in a heterogeneous but well-characterized contemporary cohort of patients with AF, most of whom were

receiving oral anticoagulation medication. Two previous analyses have evaluated the prognostic value of sNfL in predicting stroke in AF. Among 967 patients from the aspirin arm of the Atrial Fibrillation Clopidogrel Trial With Irbesartan for Prevention of Vascular Events (ACTIVE A) trial,¹⁷ a nonsignificant association between sNfL levels and risk of stroke and systemic embolism was observed, possibly reflecting the relatively small sample size and limited number of events ($n = 126$ for total stroke/systemic embolism). A larger analysis¹⁸ pooling the aspirin arms of ACTIVE A and Apixaban Versus Acetylsalicylic Acid (ASA) to Prevent Stroke in Atrial Fibrillation Patients Who Have Failed or Are Unsuitable for Vitamin K Antagonist Treatment (AVERROES) found higher sNfL to be independently associated with stroke/systemic embolism (adjusted HR, 1.27; 95% CI, 1.10-1.46; $P = .001$). Importantly, both analyses were restricted to patients unfit for oral anticoagulation medication, limiting generalizability to contemporary AF populations. With more than 90% of patients in this analysis receiving anticoagulation treatment at baseline, we show robust associations between sNfL and risk of stroke. Our analysis also suggests the association between sNfL and adverse vascular outcomes may be stronger in patients not receiving anticoagulation medication at baseline. This may indicate that sNfL's association with vascular outcomes depends on the degree of thromboinflammatory activity, which is attenuated by

anticoagulation medication. However, only a small proportion of patients in this analysis were not taking anticoagulation medication, and therefore this result should be interpreted cautiously.

We found the association between sNfL and MVE was stronger in patients without a history of stroke or TIA. Patients with prior stroke or TIA may have an already elevated baseline sNfL from large-scale neuronal damage, which may diminish the biomarker's incremental association with events due to a ceiling effect, although we cannot exclude a chance finding. Additionally, the association between increasing sNfL and MVE was consistent across different AF types (ie, paroxysmal, persistent, and permanent), suggesting that vascular stress reflected by sNfL is relevant regardless of AF chronicity or pattern. Our analysis also showed a significant association between sNfL and HF-related hospitalization, all-cause mortality, and cardiovascular mortality, with similar effect sizes found in the previous analyses of ACTIVE A¹⁷ and AVERROES.¹⁸ Our findings confirm these results in a large sample of patients who were mostly taking anticoagulation medication and had substantially longer follow-up and a larger number of outcome events. Thus, our results highlight the robustness of sNfL associations in a well-treated contemporary AF population.

Although sNfL is a highly specific marker of neuronal injury, our study demonstrated an association between elevated sNfL and systemic vascular risk. This is consistent with mechanisms postulated in prior studies. An Austrian Stroke Prevention Family Study²⁴ analysis found higher sNfL to be associated with diabetes, hypertension, and hypercholesterolemia in neurologically healthy, community-dwelling individuals. Other studies^{25,26} have shown an association between sNfL levels and inflammatory cytokines and immune-inflammation index in healthy individuals. Additionally, the observed association may point to the bidirectionality of heart-brain pathophysiology. Preclinical studies^{27,28} have shown that myocardial injury releases cardiac-derived inflammatory mediators, which can traffic to the brain and induce neuroinflammation, promoting neuronal damage that may lead to increases in sNfL. Thus, the association between sNfL and vascular outcomes in AF may be a broader marker of subclinical vascular, inflammatory, and consequently neuronal injury.⁹

Nonetheless, further studies are needed to explore the mechanistic relationships between sNfL and cardiovascular disease.

Despite the association between risk and the biomarker-based ABC-AF risk score, a prior attempt to translate biomarker-guided risk stratification into personalized management has not improved clinical outcomes and was associated with some safety concerns.²⁹ These findings do not diminish the potential importance of mechanistic analyses and risk prediction in clinical practice. Our findings suggest that sNfL may provide incremental prognostic information beyond established risk scores, such as the ABC-AF score for MVE. While effect estimates were similar for stroke, some of the *P* values for change were not statistically significant, likely at least in part due to fewer stroke events. Also, our results indicate that sNfL may reflect a different mechanistic pathway than traditional cardiac biomarkers, potentially creating a path for novel treatments in these patients at high risk of adverse outcomes despite the use of oral anticoagulation medication.^{5,20,30} This should be investigated in future studies.

Limitations

This study has limitations. Due to recruitment within Switzerland exclusively, this cohort lacks racial diversity, and the generalizability of our findings may be limited. Moreover, due to the observational nature of this study, no causal inferences can be drawn and the possibility of residual confounding despite extensive adjustment for risk factors cannot be excluded. Additionally, sNfL levels were assessed only at baseline and may have changed over the prolonged follow-up, resulting in dilution of the observed effect estimates.

Conclusions

In this cohort study of patients with AF, increasing sNfL levels were independently associated with MVEs, cardiovascular death, nonfatal stroke, HF-related hospitalization, and all-cause mortality. Serum NfL may serve as a useful noncardiac biomarker to identify patients with AF at higher risk of adverse cardiovascular outcomes.

ARTICLE INFORMATION

Accepted for Publication: March 2, 2026.

Published Online: March 29, 2026.
doi:10.1001/jamacardio.2026.0922

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Statistical analysis: Baskaran, Meyre, Coslovsky. **Obtained funding:** Kühne, Kuhle, Bonati, Osswald, Conen.

Administrative, technical, or material support: Krisai, Kühne, Aeschbacher, Kuhle, De Perna, Ammann, Sinnecker, Conte, Moutzouri, Paladini, Osswald. **Supervision:** Kühne, Ammann, Beer, Sinnecker, Conte, Conen.

Conflict of Interest Disclosures: Dr Krisai reported personal fees from Biosense Webster, Bristol Myers Squibb, and Pfizer outside the submitted work. Dr Kühne reported grants from Swiss National Science Foundation, Swiss Heart Foundation, the Foundation for Cardiovascular Research Basel, Roche, Bayer, Bristol Myers Squibb, Boston Scientific, Daiichi Sankyo, Pfizer, Abbott, and Medtronic and royalties from Springer Nature for an electrocardiogram book outside the submitted work. Dr Bonati reported grants from Swiss National Science Foundation during the conduct of the study. Dr Beer reported grants from Swiss National Foundation of Science and Swiss Heart Foundation during the conduct of the study and from Kardio Foundation Baden Switzerland outside the submitted work. Dr Moschovitis reported consulting fees, honoraria, and/or personal fees for taking part to advisory board in the field of heart failure and for presentations in the field of lipid management, hypertension, and heart failure from AstraZeneca, Boehringer Ingelheim, Daiichi Sankyo, Vifor, Gebro Pharma, Novartis, and Bayer outside the submitted work. Dr Osswald reported grants from Swiss National Science Foundation and Swiss Heart Foundation during the conduct of the study. Dr Conen reported personal fees from Abbott outside the submitted work. No other disclosures were reported.

Funding/Support: The Swiss-AF study is supported by grants from the Swiss National Science Foundation (grants 33CS30_148474, 33CS30_177520, 32473B_176178, and 32003B_197524), the Swiss Heart Foundation, the Foundation for Cardiovascular Research Basel, and the University of Basel. Dr Conen is a recipient of a Population Health Research career award.

Role of the Funder/Sponsor: The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Meeting Presentation: This paper was presented at ACC.26; March 29, 2026; New Orleans, Louisiana.

Data Sharing Statement: See Supplement 2.

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